

Contents

1	draft	1
1.1	TO_DRAFT introduction	1
1.2	paranoid & reparative reading	2
1.2.1	paranoid reading	2
1.2.2	reparative reading	4
1.3	the protocol of capture	6
1.3.1	summary	6
1.3.2	1) <i>is covetous</i>	6
1.3.3	2) <i>creates new territories for capture.</i>	9
1.3.4	3) <i>degrades resources</i>	12
1.3.5	4) <i>totalizing view</i>	15
1.3.6	5) <i>proffers a narrative of abundance</i>	18
1.4	web scraping	21
1.4.1	outline	21
1.4.2	manipulating HTTP: Heritage Foundation	21
1.4.3	the parts of a URL	24
1.4.4	bot blockers	24
1.4.5	bypassing blockers: ghost browsers	25
1.4.6	bypassing blockers: xhr	26
1.4.7	bypassing blockers: manipulating the URL	27
1.4.8	web crawling: heritage.org	29
1.5	TO_DRAFT close reading: heritage foundation	30
1.5.1	scaling small	30
1.6	TO_DRAFT conclusion: shame, sedgwick, contagion	31
1.6.1	shame as contagion	31
1.6.2	Sedgwick's critique of Butler	31

1 draft

1.1 TO_DRAFT introduction

- start with a problem: ROGD
- could do a passage from *Irresistable Damage*?
- could start with sections from the Design Justice conference paper.

1.2 paranoid & reparative reading

1.2.1 paranoid reading

Paranoid reading is an affective stance that structures critical analysis. Paranoia is first defined by Sedgwick in her introduction to a 1997 book of essays, *Novel Gazing: Queer Readings of Fiction*. In the introduction, Sedgwick explains that the essays "reflect a distance from those master-figures or master-discourses," by which she means New Historicist work that is directly indebted to "Marx, Nietzsche, and Freud" (*Novel Gazing* 2). Critical indebted to these writers, pursues a logic of demystification and exposure, which seeks to reveal the hidden realities of oppression. what Sedgwick calls. "In the context of recent U.S. critical theory," Sedgwick explains, "to apply a hermeneutic of suspicion" is, I believe, widely understood as a mandatory injunction rather than a possibility among other possibilities" (*Novel Gazing* 3).

There are five qualities that define paranoia, according to Sedgwick. Paranoia:

- is anticipatory. It predicts and preempts every possibility, as Sedgwick says, "The first imperative of paranoia is 'There must be no bad surprises'" (emphasis original; *Novel Gazing* 9). Rather than accept unfavorable possibilities, according to the "future-oriented vigilance [that] paranoia generates... bad news [must] be always already known" (*Novel Gazing* 10).
- is reflexive, mimetic. Paranoia reproduces itself, propagating and spreading like a virus from host to host. Sedgwick explains that "Paranoia seems to require to be imitated in order to be understood... 'Anything you can do, I can do worse' and 'Anything you can do, I can do first'" (emphasis original *Novel Gazing* 10). One effect is that it creates self-reinforcing oppositional relationships, such as Foucault's posing of repressive hypothesis, which leads to a self-perpetuating narrative of hegemony vs subversion that, according to Sedgwick "propagat[es] the repressive hypothesis ever more broadly by means of displacement, multiplication, hypostatization" (*Touching Feeling* 10).
- is strong theory - it invades, consumes all phenomena into its worldview. This quality gives paranoia "reach and reductiveness," which makes "tautological thinking hard to identify, even as it makes it compelling and near-inevitable" (*Novel Gazing* 13, 15). For example, referring to D.A. Miller's book, *The Novel and the Police*, a study of surveil-

lance, policing, and disciplinary culture in Victorian novels, Sedgwick points out that "the main argument or 'strong theory' of *The Novel and the Police* is entirely circular: everything can be understood as an aspect of the carceral, therefore the carceral is everywhere" (*Novel Gazing* 14).

- is negative theory - paranoia is driven by the minimization of negative affect rather than the maximization of positive affect, "the anticipation of pain in one case, [against] the provision of pleasure in the other" (*Novel Gazing* 16). Sedgwick poses Silvan Thompkin and Melanie Klein's positive affect theories against "paranoid Freudian epistemology," where "it is implausible enough to suppose then that truth could be even an accidental occasion of joy; inconceivable to imagine joy as a guarantor of truth" (*Novel Gazing* 16).
- places its faith in exposure - its obsessive goal is to unearth, surface, reveal some hidden reality (often of oppression or injustice) as knowledge. It is, Sedgwick explains, "Like the deinstitutionalized person on the street who, betrayed and plotted against by everyone else in the city, still urges on you the finger-worn dossier bristling with his precious correspondence" (*Novel Gazing* 17). This faith "acts as though its work would be accomplished if only it could finally, this time, somehow get its story truly known" (17).

For Sedgwick, paranoid reading is a crucial framework for analysis, allowing much important work to happen. She refers to her own work and how it has been driven by paranoia. For example, her field-defining work, *The Epistemology of the Closet*, is a study in hidden sexualities, and the power structures they maintain, in literature. Sedgwick's agenda in the book is to expose the dependence of a privileged heterosexual position upon the existence of a subordinated homosexual. How do these sexual categories and closet which they structure appear in 19th-20th century literature? For example, her chapter on Oscar Wilde's *The Picture of Dorian Gray* examines abstraction and figuration as strategies for encoding male on male desire which otherwise couldn't be expressed: "the modernist impulse toward abstraction in the first place owes an incalculable part of its energy precisely to the turn-of-the-century male homo/hetero definitional panic. . . the 'figuration' that had to be abjected from modernist self-reflexive abstraction was not the figuration of figurality itself, but, rather, that represented in a very particular body, the desired male body" (*Epistemology*, 167). This figuration

I turn to in my third chapter.¹

Nonetheless, these structures are unhelpful for thought, even "a cognitive danger... a moralistic tautology that became increasingly incabable of recognizing itself as such" (*Touching Feeling* 12). Reading Foucault's *History of Sexuality, Vol. 1*, Sedgwick says, "I knew what I wanted from it: some ways of understanding human desire that might be quite to the side of prohibition and repression, that might hence be structured quite differently from the heroic, 'liberatory', inescapably dualistic righteousness of hunting down and attacking prohibition/repression in all its chameleonic guises" (*Touching Feeling* 10).

1.2.2 reparative reading

She thus pursues a critical method outside the paranoid logic, which she calls "reparative reading." In contrast to paranoid reading, reparative reading prioritizes pleasure, surprise, and hope:

to read from a reparative position is to surrender the knowing, anxious paranoid determination that no horror, however apparently unthinkable, shall ever come to the reader as new: to a reparatively positioned reader, it can seem realistic and necessary to experience surprise. Because there can be terrible surprises, however, there can also be good ones. Hope, often a fracturing, even a traumatic thing to experience, is among the energies by which the reparatively positioned reader tries to organize the fragments and part-objects she encounters or creates. Because she has room to realize that the future may be different from the present, it is also possible for her to entertain such profoundly painful, profoundly relieving, ethically crucial possibilities as that the past, in turn, could have happened differently from the way it actually did. (/Novel Gazing 23-24)

¹Yet, even in this early work, Sedgwick pursues a critical method that is local. In *Epistemology*, she writes that: "A point of the book is not to know how far its insights and projects are generalizable... The book aims to resist in every way it can the deadening pretended knowingness by which the chisel of modern homo/heterosexual definitional crisis tends, in public discourse, to be hammered most fatally home" (12). Here she gives away a certain critical openness that will then lead her to her thinking in *Novel Gazing* and *Touching/Feeling*. She isn't aiming for generalizable or abstract critical theories—she knows that any totalizing theory would be more hurtful than helpful. She's carving out a space for what she eventually comes to say about the need to open up possibilities for connection with texts.

Reparative motives do not exist in a vacuum from paranoia. Rather, the both often work together: "Powerful reparative practices... infuse self-avowedly paranoid critical projects" (*Novel Gazing* 8). However, a reparative recognizes most of all "that the culture surrounding it is inadequate or inimical to its nurture" (*Novel Gazing* 28). In response, Sedgwick argues, is "additive and accretive... it wants to assemble and confer plenitude on an object that will then have resources to offer an inchoate self" (*Novel Gazing* 26-27).

Close reading is the tool par excellence of reparative work. Particularly, Sedgwick stipulates a kind of "unhurried, undefensive, theoretically galvanized practice of close reading," one which permeates the essays in *Novel Gazing*. These essays, according to Sedgwick, approach criticism in "a much more speculative, superstitious, and methodologically adventurous state where recognitions, pleasures, and discoveries seep in only from the most stretched and ragged edges of one's competence" (*Novel Gazing* 3).

Often these readings begin from or move toward sites of same-sex, interpersonal eroticism—but not necessarily so. It seems to me that an often quiet, but very palpable presiding image here—a kind of *genius loci* for queer reading—is the interpretive absorption of the child or adolescent whose sense of personal queerness may or may not (yet?) have resolved into a sexual specificity of proscribed object choice, aim, site, or identification. Such a child—if she reads at all—is reading for important news about herself, without knowing what form that news will take; with only the patchiest familiarity with its codes; without, even, more than hungrily hypothesizing to what questions this news may proffer an answer. *Novel Gazing* 2-3

Years later, Sedgwick will perform her own reparative reading of shame, an affect which is typically a negative, structuring force in queer identity. In the essay entitled "Shame, Theatricality, Queer Performativity," from her collection of essays in *Touching Feeling*, Sedgwick examines the writings of Henry James for evidence of shame. However, rather than plumb shame's depths for what it reveals about a hidden sexuality, Sedgwick explores how it might function as a generative resource for creativity. She uses shame, which she associates with words like "blusing" and "flushing," it to pull other affects and images into relation. For example, when reflecting on his writing practice, James,

describes his revision of the early fictions both as his (or their?) way of "remaining *unshamed*" and as a process by which they

have "all joyously and *blushingly* renewed themselves" (345; emphasis added). What James seems to want here is to remove the blush from its terminal place as the betraying blazon of a ruptured narcissistic circuit, and instead to put it in circulation: as the sign of a tenderly strengthened and indeed now "irresistible" bond between the writer of the present and the abashed writer of the past, or between either of them and the queer little *conceptus*. (emphasis original; *Touching Feeling* 41)

Using shame as a connective tissue, Sedgwick moves from "blushing" and "flushing" to other associated terms, like "fond." This term, which Sedgwick describes as "one of James's most cherished words," "marks the place of the author's pleasure in dramatizing himself as all but flooded with self-absorbed delusion and embarrassment, but equally with pleasure" (*Touching Feeling* 55). From James's *fondness* with his self-embarrassment, Sedgwick also reads a fixation with anal metaphors—for, as Sedgwick points out, "fond is also the French word for bottom, [which] may explain its affinity with [other terms like] "retrospect," the "backward view," even with the "thrilling tale" (*Touching Feeling* 55). Moving from blushing to fond, then, enables Sedgwick to discover "a node where the theatrics of shame, affection, and display are brought together with a compositional principle and at the same time lodged firmly, at the level of the signifier, in a particular zone of the eroticized body" (*Touching Feeling* 56).

1.3 the protocol of capture

to do:

- change the main argument of each sub point to address the way it affects meaning.

1.3.1 summary

Text capture is the process in which text data is extracted from web servers.

This kind of capture follows the same protocols as paranoia, driven by a totalizing impulse which seeks to incorporate everything that it can.

There are five central principles to text capture, which map loosely to Sedgwick's principles of paranoia. Capture, in turn,

1.3.2 1) *is covetous*

- it desires more language

If paranoia is anticipatory, capture is covetous. It always wants more, propelled by a seeming necessity to aggregate resources into its hold.

Ulises A. Mejias and Nick Couldry describe the acceleration of of electronic information harvesting as a "data grab." They compare it to traditional colonialism's "land grab," the long process over several centuries where Europe seized over 84 percent of the world, despite representing only 8 percent of its landmass (Mejias & Couldry 5). As they explain, "[B]oth systems are founded on an appropriation of the world's resources, treating them as 'just there' for the taking. Historically it was the territories... In the contemporary era, it is human life, extracted in the form of data" (Mejias & Couldry 18). Just as colonizers allocated lands not represented in their European maps for the Crown, so now tech companies take all of the data they can get.

The capture of text, image, videos from the websites as unclaimed resources often occurs through methods such as web scraping. According to Cloudflare, a major tech provider for cloud and cyber computing services, AI crawlers increased sharply, ranging between 18%-48% in the months between May 2024 and May 2025. Google's "Googlebot" maintained its prominence as the largest web scraper, the highest increases came from "GPTBot" and "META-ExternalAgent," which "highlight[s] the increasing prominence of OpenAI and Meta in this category."²

As researcher Ramya Chandrasekhar points out, "certain well-resourced actors like Big Tech are able to take advantage of data flows on the open web to internet to train proprietary foundation models, while giving little to no value back to either the maintenance of shared informational resources or communities of commoners" (Chandrasekhar 2).³

In tech industry terms, automated web scraping is inevitable. An annual report by one data company, Zyte, reports as much:

For years, we have talked about the "cat-and-mouse" between data- gatherers and data owners. But 2025 marked a significant transition in this space. The pace of updates from bot management vendors has accelerated, and the old methods of maximizing data success are becoming less effective.

²"From Googlebot to GPTBot: Who's Crawling Your Site in 2025." The Cloudflare Blog, 1 July 2025, <https://blog.cloudflare.com/from-googlebot-to-gptbot-whos-crawling-your-site-in-2025/>.

³Chandrasekhar, Ramya. Legal Frictions for Data Openness: Reflections from a Case-Study on Re-Use of the Open Web for AI Training. Zenodo, 2025, p. 75. HAL Archives Ouvertes, <https://doi.org/10.5281/zenodo.15097649>.

This is a good thing.

The complexity of modern anti-bot measures has gone past the point where human developers can, or should, manage it manually.

...

The friction of the past 15 years is giving way to a future where data extraction is seamless, compliant, and powered by intelligent agents. (5-6) ⁴

With turns of phrase that evoke the thin confidence of generative AI, the report convinces the reader that, indeed, more barriers to web scraping "updates from bot management vendors" actually spells opportunity: a future where everything is automated. Where there was friction, now it is seamless, with the help of "intelligent agents." The use of vague terms, like "maximizing data success", paper over the goal of vacuuming as much data as possible. For this, and undoubtedly many, companies, increases in automation means it's the time to ramp up production.

More data means more training examples, which results in higher-quality systems that make better predictions.⁵ Unfortunately, as AI Ethics researchers have been pointing out for years, more data means more problems. Their prescient paper, released the year before ChatGPT came out, "The Dangers of Stochastic Parrots: Can Language Models Be Too Big?" (2021), posed the problem directly in the title: *Can language models be too big?*⁶ The answer, it turns out, is yes, with large model sizes (even in 2021, the era of GPT-3), creating issues with environmental and data sustainability, discussed in more detail in point #3 below.

Four years and not a few GPTs later, Gael Varoquaux, Alexandra Sasha Luccioni and Meredith Whittaker furthered this research into model performance itself. Drawn by the "bigger-is-better paradigm," LLMs are now so big, they are even plateauing in performance (a point addressed in principle #3, that capture diminishes resources and degrades quality).⁷

⁴Zyte. 2026 Web Scraping Industry Report - PDF. <https://www.zyte.com/whitepaper-ebook/2026-web-scraping-industry-report/>. Accessed 8 May 2026.

⁵Indeed, the power of scale is well documented in Machine Learning research. Kaplan, Jared, Sam McCandlish, Tom Henighan, et al. "Scaling Laws for Neural Language Models." arXiv:2001.08361. Preprint, arXiv, January 23, 2020. <https://doi.org/10.48550/arXiv.2001.08361>.

⁶After she refused to retract her affiliation from the publication, Timnit Gebru was famously fired from Google. Hao, Karen (4 December 2020). "We read the paper that forced Timnit Gebru out of Google. Here's what it says". MIT Technology Review.

⁷Alexandra Sasha Luccioni is a research scientist at HuggingFace and head of Sustain-

1.3.3 2) *creates new territories for capture.*

- it opens new domains of extraction from language
- the contract effectively creates a new territory for capture, where there hadn't been one before
 - the TOS is a formalization of capture.
 - Shoshana Zuboff's point on new contractual forms.
- a new form that emerges with web scraping: content providers are finding new ways to profit from their ownership of labor. -> could this then lead to the new transformative use argument?

Besides the desire for more, capture creates new contractual forms which formalize ever more territories for capture. One might say that the pretenses required for capture automates the creation of new territories, which relates to paranoia insofar as paranoia is mimetic, automatically reproducing itself through certain performative strategies: "as a locus of reflexive mimeticism," Sedgwick says, "paranoia is nothing if not teachable" (*Novel Gazing* 15). Like a good student imitates her teacher, it is a self-replicating phenomenon, one that "would seem to grow like a crystal in a hypersaturated solution, blotting out any sense of the possibility of alternative ways of understanding or things to understand" (*Novel Gazing* 10-11).

One of these self-perpetuating, contractual forms, Mejias and Couldry explain, is the Terms of Service, or TOS. Mejias and Couldry compare the TOS to the *Requerimiento*, an explicitly coercive document that the Spanish colonizers would read to Indigenous peoples of Americas as they entered their land and villages to take its resources and people under control of the Spanish Crown. Similarly, the TOS also makes claims upon the user. The language of the TOS isn't as as violent or even explicitly coercive as the *Requerimiento*, which declared that any those who resisted would be violently subdued. Nonetheless, Mejias and Couldry point out, they don't need to be, for "without most of us ever having read those original terms and conditions, we have reorganised our lives around whatever rules Google imposes on us" (12). For example, the 2007 version of the TOS asserts that:

able AI Group. Meredith Whittaker is the president of the Signal Foundation, former faculty director & co founder of the AI Now Institute, and was employed at Google for 13 years, where she was a core organizer of the Google Walkouts. Bloomberg. "Google Protest Leader Leaves, Warns of Company's Unchecked Power". July 18, 2019. Retrieved July 18, 2019.

You give Google a perpetual, irrevocable, worldwide, royalty-free, and non-exclusive license to reproduce, adapt, modify, translate, publish, publicly perform, publicly display and distribute any Content which you [the user] submit, post or display on or through, the Services. (Rpt. in Mejias & Couldry 12).

More recently, the wording of TOS has muted further the explicitness of the data grab. The more recent strategy has shifted from assertive to excessive and at times obscure language (these TOS now occupy many pages). For example, the 2024 Google TOS, which is now held on its own subdomain, <https://policies.google.com>, phrases like "operating and improving the services" and "to customize our services for you" offer the necessary cover for "using automated systems and algorithms to analyze your content" ("License", "Google Privacy and Terms"). Additionally, the TOS contains strange and difficult to parse formulations, like "We need your permission if your intellectual property rights restrict our use of your content. You provide Google with that permission through this license" ("License", "Google Privacy and Terms"). The meaning appears to be that Google will need, yet already has obtained, permission from the user. But what exactly is one giving permission for Google to do? The TOS does not say.

That is, if one even bothers to read the terms of service. Whereas the *Requerimiento* relied on using inaccessible language (As Mejias and Couldry point out, "[the conquerors] read this document in Spanish. . . to indigenous or First Nation peoples who did not speak Spanish!" (13)), more recent TOS rely on the user not even checking the terms. In other words, both approaches rely on the user not knowing the extent of the rights being taken from them.

Contractual forms, as preciently pointed out by Shoshana Zuboff in 2015, is one of the key innovations of what Zuboff calls the "logic of accumulation." Zuboff analyzes an internal document written by Google's chief economist, Hal Varian, which introduces four new categories of "computer-mediated economic transactions," including "data extraction and analysis," "new contractual forms," "personalization and customization," and "continuous experiments" (Zuboff 78). Most of these, Varian explains, rely on the nearly infinite streams of data that platforms can capture from users, and on the fact that these streams are mostly hidden from view, thus creating asymmetrical relationships between the consumer and the business which, happily for Google, obviates the role of trust. As Zuboff explains, "New possibilities of subjugation are produced as this innovative institutional logic thrives on unexpected and illegible mechanisms of extraction and control that exile

persons from their own behavior" (85).

While, I argue, the logic of accumulation and surveillance capitalism has existed for centuries, there is an emergent effect which seems specific to digital contexts.⁸ Zuboff describes this effect with the term "informate":

when it comes to information technology, automation simultaneously generates information that provides a deeper level of transparency to activities that had been either partially or completely opaque. It not only imposes information (in the form of programmed instructions), but it also produces information. . . This produces action linked to a reflexive voice, as computer mediation symbolically renders events, objects, and processes that become visible, knowable, and shareable in a new way. (76)

"Informate" describes this tendency of computer mediated action to create byproducts of information that were previously inaccessible, even unimaginable.⁹ According to Zuboff, this tendency turns all digital data "electronic text"—text which is ripe for extraction via reading. When computers automate action, they create new texts, like foam rising from a boiling pot.

With regard to web scraping in particular, one emergent territory is the "permission-based model." Cloudflare, for example, has employed this model where bots must pay for seamless access to certain domains. A blog post frames this model as a logical step in the information exchange ecology which is the internet:

For decades, the Internet has operated on a simple exchange: search engines index content and direct users back to original websites, generating traffic and ad revenue for websites of all sizes. This cycle rewards creators that produce quality content with money and a following, while helping users discover new and

⁸Indeed, slavery, as an economic order, could be described in the same terms that Zuboff uses to describe surveillance capitalism: "A parasitic economic logic in which the production of goods and services is subordinated to a new global architecture of behavioral modification" (from book, *The Age of Surveillance Capitalism*). Brown's work, for example, surfaces the quantifying and policing practices that Black people have endured since slave ships. And more recently, as Benjamin argues, "tech advances are sold as morally superior," although "they could not exist without data produced through histories of exclusion and discrimination" (5).

⁹Zuboff's ideas have been percolating for some time. Philip Agre's 1994 paper, "Surveillance And Capture: Two Models Of Privacy," details similar views on computation and surveillance, as well as concepts, such as Agre's "grammar of action," by which a human activity is made computable.

relevant information. That model is now broken. AI crawlers collect content like text, articles, and images to generate answers, without sending visitors to the original source – depriving content creators of revenue, and the satisfaction of knowing someone is viewing their content. (“Cloudflare Just Changed How AI Crawlers Scrape the Internet-at-Large; Permission-Based Approach Makes Way for A New Business Model”)

While large-scale web crawling originated to help populate and search engines, which was legitimized as "fair use" in a famous copyright lawsuit against Google, *The Authors Guild Inc., et al. v. Google, Inc.* (2015), Cloudflare points out that today's scrapers and crawlers do not direct users back to the original source—rather, the AI training process *severs* the data completely from its context, including its source. Text is siphoned from web pages and compiled into massive collections which are fed, bit by bit, into a machine learning algorithm. As the Cloudflare blog post points out, the severed connection to the source cannot be monetized, "depriving content creators of revenue... and satisfaction."

As a solution, Cloudflare is now offering the "permission based model," where companies must pay content creators for access to their web data. If the scraper does not have explicit permission to retrieve data, then they will be blocked. The announcement of Cloudflare's new policy came with written endorsement from publishers and media outlets. Roger Lynch, CEO of media conglomerate Condé Nast, which owns *Vogue*, *The New Yorker*, *Architectural Digest*, *Vanity Fair*, *Pitchfork*, *Wired*, *Bon Appétit*, *Ars Technica*, among others, declares that, "Cloudflare's innovative approach to block AI crawlers ... is a critical step toward creating a fair value exchange on the Internet that protects creators, supports quality journalism and holds AI companies accountable" (“Cloudflare Just Changed How AI Crawlers Scrape the Internet-at-Large; Permission-Based Approach Makes Way for A New Business Model”). Of course, there is no guarantee (or even expectation) that such payments would reach the pockets of actual content creators, authors, copywriters, artists, who are populating these sites with their work. If the new policy rewards someone, it's not the original creator but the publisher who enters into agreements with the tech companies. In that sense, the "fair value exchange" cited by Lynch is in fact another way of further monetizing what is already an extractive practice of capture.

1.3.4 3) *degrades resources*

- it **degrades the quality** of language

The third principle is one of degradation—not only of the resources used up in the extraction process but also of that which is produced by it, that is, the product of capture. This principle takes its inspiration from Sedgwick’s theorization of paranoia as a negative affect. Sedgwick explains that although positive and negative affects can be similarly constituted, in that they may both be based on pessimism (“they are driven toward different goals: “the anticipation of pain in one case, the provision of pleasure in the other” (*Novel Gazing* 16). Crucially, the anticipation of pain, for Sedgwick, is total: “the mushrooming, self-confirming strength of a monopolistic strategy of anticipating negative affect can have... the effect of entirely blocking the potentially operative goal of seeking positive affect” (*Novel Gazing* 15). Negative theory, then, is all consuming, with the effect of blocking out other kinds of relation. (Comparing against Freud, remarks about a reading of Proust) For someone driven by negative theory, “it is implausible enough to suppose then that truth could be even an accidental occasion of joy; inconceivable to imagine joy as a guarantor of truth” (*Novel Gazing* 16).

I argue that, in seeking to maximize profit, capture follows a negative principle of degradation across environmental, material, labor, and even production contexts. First, and most obviously, the cycle of capture depletes natural resources. Processing data at large scales requires equally large processing infrastructures like data centers. Not only do these centers exhaust precious water in order to cool them, their disproportionate use of energy most visibly results in rising prices for electricity across the country, and perhaps less visibly (though more devastating) in its effects on the environment.¹⁰

Beyond energy consumption, acquiring the basic materials for data processing, like the minerals used to make computer hardware, involves invasive and exploitative mining practices. Much of this mining is occurring in the southern hemisphere, in particular the African continent. The harmful effects include toxifying the environment, particularly the water supply, and intensifying social upheaval by contributing to resource conflicts.¹¹ More-

¹⁰Nguyen, Terry. WHAT HAPPENS WHEN DATA CENTERS COME TO TOWN? U of Michigan. July 2025; Mineo, Liz. “Why Are Communities Pushing Back against Data Centers?” Harvard Gazette, April 9, 2026. <https://news.harvard.edu/gazette/story/2026/04/why-are-communities-pushing-back-against-data-centers/>.

¹¹Vulević, Tijana, Nada Dragović, Lazar Radulović, et al. ANALYSIS OF SOCIAL IMPACTS OF MINES AND PATHWAYS TO RESPONSIBLE MINING; Leonard, Llewellyn. “Socio-Environmental Impacts of Mineral Mining and Conflicts in Southern and West Africa: Navigating Reflexive Governance for Environmental Justice.” Environmental Research Letters 19, no. 10 (2024): 104013. <https://doi.org/10.1088/1748-9326/ad7047>.

over, these troubling reports are hard to find among the decidedly more positive ones from institutions such as the World Bank and McKinsey, which emphasize Africa's "rich and diverse mineral endowment, strong discovery potential, and lower exploration costs."¹²

The human labor required to process this data also tells a devastating story. The development of automated tools requires high levels of human intervention, what is called "data annotation" or "data work." Human labor is necessary to prepare the data used for training an ML model that will then be used to produce a tool. For example, a computer vision tool that tags images with descriptive information requires training from a dataset of images that are correctly mapped to existing tags. Humans are hired, usually on a contractor basis, to manually apply the tags to generate the dataset.

Like other kinds of gig work, data annotation exploits laborers. Much of the work involves repeating the same task, over and over again, such as applying certain categories to hundreds or thousands of images. For this kind of labor, the pay is low, and usually sourced from countries that do not have strict labor law. Additionally, the psychological effects can be severe. For example, OpenAI hired contractors from Kenya to train its GPT models—specifically, to remove toxicity and violence from any model outputs. They tasked the workers with labelling any instances of violence and abuse in the model. As Billy Perrigo reports, the company paid the workers less than \$2 per hour to view and flag the violent content.¹³ They were not offered mental health services. As Berhan Taye points out, although these workers are hired from companies like OpenAI, Google and Meta, they do not have the same perks as regular employees of these companies do.¹⁴

Degradation spreads from the material used for training to the products of training themselves. This effect is directly a function of scale, which effect models in two ways: model plateau and model collapse. First, model plateau shows that large ML models, like LLMs, apparently have an upper limit, plateauing in performance after they reach a certain size. Exploring

¹²The World Bank. "Mineral Resources of Africa." Accessed September 14, 2026. <https://www.worldbank.org/en/brief/2025/10/15/mineral-resources-of-africa>; McKinsey. "Critical Minerals in Africa: Advancing Mining Excellence." Accessed September 14, 2026. <https://www.mckinsey.com/industries/metals-and-mining/our-insights/african-bedrock-how-the-continents-minerals-can-help-power-the-world>.

¹³Perrigo, Billy. "Exclusive: The \$2 Per Hour Workers Who Made ChatGPT Safer." TIME. 18 Jan. 2023, <https://time.com/6247678/openai-chatgpt-kenya-workers/>.

¹⁴Berhan Taye. "The Plight of AI Production Pipeline Workers." Global Humanities Institute 2024, *Design Justice AI* June 30 - July 13, 2024. Pretoria, South Africa. <https://criticalai.org/2025/11/28/program/>

what they term the "bigger-is-better paradigm", Varoquaux et al. note that "the narrative that the larger the AI system the more valuable, powerful and interesting it is is increasingly seen as common sense" (Varoquaux et al. 1). However, their experiments show the opposite, that "scale is not well correlated with better performance in certain contexts, and in fact benefits of scale tend to saturate" (Varoquaux et al. 2).

There is also the degradation of the material used for training—the training data itself. As more AI-generated content proliferates on the internet, we approach what Matt Kirschenbaum calls the "textpocalypse," where text that is generated from an ML model dominates the content on the internet.¹⁵ Interestingly, researchers have found that this kind of text corrupts model training, a phenomenon which they call "model collapse." Model collapse describes what happens to a model when it is trained and re-trained on ML-generated data. As the researchers explain, is "a degenerative process... in which the data they [the models] generate end up polluting the training set of the next generation" (Shumailov et al 755).¹⁶

The data that is generated by AI will be fed to AI creates kind of digital ouroboros. The ouroboros, the snake eating its own tale, is an ancient symbol of the eternal cycle of life and death, of renewal. However, in this version, the cycle is not eternal, the snake ends by totally consuming himself.

1.3.5 4) *totalizing view*

- it **collapses** language into one meaning.

Like paranoia, capture is *strong theory*. Drawing again from Silvan Tompkins, Sedgwick characterizes strong theory by its "reach and reductiveness," and by the "size and topology of the domain that it organizes" (*Novel Gazing* 13).

The forces of capture converge toward centralization and total control. They are against dispersion. They prefer hegemonic structures, for example of signification, where there can be only one meaning. The hegemony of meaning by necessity conscripts the security enterprise into the commercial one.

¹⁵Kirschenbaum, Matthew. "Prepare for the Textpocalypse." *The Atlantic*, 8 Mar. 2023, <https://www.theatlantic.com/technology/archive/2023/03/ai-chatgpt-writing-language-models/673318/>. Technology.

¹⁶Shumailov, Ilia, et al. "AI Models Collapse When Trained on Recursively Generated Data." *Nature*, vol. 631, no. 8022, July 2024, pp. 755–59. [www.nature.com, https://doi.org/10.1038/s41586-024-07566-y](https://doi.org/10.1038/s41586-024-07566-y).

When happens to individual words, a single word will contain contradiction. In 2006, Wendy Chun writes about this effect at the start of the fiber optic age, that "“a technology, which thrives on control, has been accepted, however briefly as a mass medium of freedom” (*Control and Freedom* vii). Chun traces this history back to the 1996 Telecommunications Act, signed into law by President Bill Clinton, which deregulated the telecommunications market, opening it up to private companies. While this move enabled the expansion of network access, it also quickly concentrated the industry into the hands of a few powerful companies. One effect, key for Chun’s argument, is the conflation of "freedom" and "control." Although fiber optic technology promises freedom of communication, it relies on strict protocols that determine how one node connects to another. To Chun, the issue isn’t the control inherent to network protocols that drive the internet, but in the way that the protocols are obscured by a narrative of freedom, which includes values of privacy: “If you believe that your communications are private, it is because software corporations, as they relentlessly code and circulate you, tell you that you are behind, and not in front of, the window” (22).

Similar conflations of "freedom" and "control" occur in the present day in language around AI production. In a proposal of recommendations for developing technology policy, OpenAI, for example, poses American values of "freedom" and "democracy" against the "autocratic, authoritarian" People’s Republic of China (PRC).¹⁷ OpenAI asserts a series of "freedoms" endows to Americans, including "the freedom to access and benefit from AGI"—that is, for consumer access to commercial products like ChatGPT; the "freedom to innovate," whereby American companies would be exempt from complying with "overly burdensome state laws" around AI use, such as those requiring safety and bias checks ("AI Laws and Regulation in the United States (2026)"); and finally, there is a "freedom to learn," which, rather slyly, refers the freedom of machines (rather than human beings) to learn (Lehane 2). The freedom to learn, as the proposal itself admits, is really the "ability to learn from copyrighted material" without having to pay for the content (Lehane 5).

The remarkable hypocrisy around data theft comes into the starkest illustration when discussing the "fair use" exemption from copyright law.

The rapid advances seen with the PRC’s DeepSeek, among other recent developments, show that America’s lead on frontier AI is far from guaranteed. Given concerted state support for critical

¹⁷On March 13, 2025, the company OpenAI responded to the call for AI policy recommendations from the White House Office of Science and Technology Policy (OSTP).

industries and infrastructure projects, there's little doubt that the PRC's AI developers will enjoy unfettered access to data—including copyrighted data—that will improve their models. If the PRC's developers have unfettered access to data and American companies are left without fair use access, the race for AI is effectively over. America loses, as does the success of democratic AI. (Lehane 9-10)

While the US's access to the same data is "fair," China's access to the same data is "unfettered"—despite there being no difference in the datasets (something like the entire web) and the methods of web scraping and web crawling, which are used to obtain them.

"Freedom" is strange in this context — it's a US value, being contrasted to China's "autocratic" values. Yet, freedom is deployed in the service of making the US policy effectively more like that of China's. The autocratic qualities of the PRC, which allows it to develop its AI free of regulation and copyright considerations, is exactly what the US is asking for in this proposal.

This effect of meaning collapse, between freedom and control, or freedom and autocracy, also occurs as a function of *scale* in capture. At large scales of text processing, what is statistically common will be amplified—that is, what is overrepresented becomes hegemonic. "Size doesn't guarantee diversity," Bender et al. explain (613). Even large collections of data, like the Colossal Cleaned Crawled Corpus (C4), which is one of the most common training datasets for LLMs, overrepresents dominant viewpoints—and not just because these viewpoints are more likely to be represented in internet spaces. In a study of the C4 dataset, Dodge et al (2021) find that the filtering of bias and toxicity about minority groups has an attendant effect of removing these perspectives entirely:¹⁸

we find that mentions of sexual orientations (lesbian, gay, heterosexual, homosexual, bisexual) have the highest likelihood of being filtered out... Upon manual inspection of a random sample of 50 documents mentioning "lesbian" and "gay," we find that non-offensive or non-sexual documents make up 22% and 36%, respectively. (Dodge et al, 2021, p. 1292)

¹⁸Dodge, Jesse, Maarten Sap, Ana Marasović, et al. "Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus." In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, edited by Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih. Association for Computational Linguistics, 2021. <https://doi.org/10.18653/v1/2021.emnlp-main.98>.

The prioritization of hegemonic viewpoints also occurs in other steps of the data gathering process, from the internet spaces themselves, like Reddit.com, which is dominated by young and male users, to the moderation of these platforms, to efficient filtering mechanisms that will inadvertently take out more information relating to minority identities than majority ones. Each step of the data gathering process, "from initial participation in Internet fora, to continued presence there, to the collection and finally the filtering of training data," Bender et al. explain, current practice privileges the hegemonic viewpoint" (614).

Not only data gathering, but data training also has the effect of reinforcing majority viewpoints. Abeba Birhane et al. (2023) write about this "Dark Side of Dataset Scaling," where ML models exhibit more bias as training datasets scale up in size. For example, the researchers' experiments find that models are more likely to categorize images of Black and Latino men with the label "criminal," (Birhane et al, 2023, p. 1231). Aligning their work in Black studies, critical data studies, and critical race scholarship, situate these results as part of a narrative of historical discrimination, which stretches from the institution of slavery to the rise of what they describe as the "for-profit prison industrial complex" (Birhane et al 2023, 1235).

This unintuitive reality, that more data results in less diverse viewpoints, which in turn results in more biased models, has a specific implication on language and language meaning. It means that the convergence of language meaning is a function of scale. The larger the scale of the dataset perhaps creates a situation where meaning whittles to a single value, like Black -> criminal.

1.3.6 5) *proffers a narrative of abundance*

- despite the decay of resources, there is a narrative of abundance.
 - this takes out one expressive materiality of language and exchanges it for a commercial materiality.
- "paranoia [capture]. . . acts as though its work would be accomplished if it could [would] finally, this time, somehow get its story known [distribute its riches]" (*Novel Gazing* 17).
- transformative use - modeling language itself

Sedgwick explains that the final quality of paranoia is that it *places its faith in exposure*. She writes, "paranoia is characterized by placing, in practice, an extraordinary stress on the efficacy of knowledge per se—knowledge

in the form of exposure. Maybe that's why paranoid knowing is so inescapably narrative" (*Novel Gazing* 17).

If paranoia places its faith in knowledge, capture places its faith in abundance. Most obviously, this faith in abundance is evident in the language of the tech CEOs themselves, in the written statements and spoken interviews. Even statements that seem to have differing messages, such as Sam Altman's "Abundant Intelligences," Mark Zuckerberg's "The Future Is For Everyone," which encourage growth at top speed, and Dario Amodei's "We Must Pace the Frontier," which encourage moderation in growth, ultimately make the same promise: a world which is abundant for everyone.^{19, 20}

Part of this abundance is a certain perspective on language and its value.

Right now, there are over 40 ongoing copyright lawsuits brought by content creators against companies like OpenAI, Anthropic, Midjourney, and many others. To defend themselves, these companies argue that their data harvesting constitutes "fair use," which is a copyright protection that allows the taking of original work depending the purpose and nature of the use, and the effect on the market for the original work, among factors.

For example, in the lawsuit brought by the NYTimes against OpenAI, OpenAI defends itself by saying they are using copyrighted data to "learn facts" about language, which is a purpose that is protected under "fair use." They claim that,

... it is fair use under copyright law to use publicly accessible content to train generative AI models to learn about language, grammar, and syntax, and to understand the facts that constitute humans' collective knowledge. OpenAI and the other defendants in these lawsuits will ultimately prevail because no one—not even the New York Times—gets to monopolize facts or the rules of language. MEMORANDUM OF LAW filed by OpenAI, February 2024

This phrase, the "facts and rules of language", and the claim that this is all that language models extract from data during the training process.

They appeal to a stipulation from the Fair Use clause, which emphasizes the importance of "transformative-ness", that is, how much the derivative

¹⁹Altman, Sam. "Abundant Intelligence." September 23, 2025. <https://blog.samaltman.com/abundant-intelligence>. Zuckerberg, Mark. "The Future is for Everyone: The Path to a Positive AI Future." August 10, 2026. <https://www.meta.com/thefutureisforeveryone/>; Dario Amodei, We Must Pace the Frontier, September 2026: <https://darioamodei.com/post/we-must-pace-the-frontier>

²⁰Like partisan politics, the top competitors in the tech industry are two sides of the same coin, only seem to come into conflict in order to stifle competition from below.

object changes from the original. They cite a legal case from 2014, *Authors Guild v. HathiTrust*, which is about the legality of search results.

we conclude that the creation of full-text searchable database is a quintessentially transformative use... the result of a word search is different in purpose, character, expression, meaning, and message from the page (and the book) from which it is drawn. Indeed, we can discern little or no resemblance between the original text and the results of the HDL full-text search. (*Authors Guild v HathiTrust* 2014, page 18)

The ruling for this case finds that search results are fundamentally transformative from the works that they reference, because they offer a new object, that is, *information about works* in their database. This ruling determines that search results are significantly different from the original.

OpenAI applies this understanding of "transformative" to defend the legality of their language models. They argue that, like search results, language models constitute a new kind of object, one that *generalize language patterns*. They explain that,

"By learning patterns from its training corpus, an AI system can eventually generate media that shares some commonalities with works in the corpus (in the same way that English sentences share some commonalities with each other by sharing a common grammar and vocabulary) but cannot be found in it." ("Comment", 9-10)

Leaving aside that while grammar and vocabulary are not copyrighted, dictionaries and grammar books are certainly copyrighted material, the use of the term "patterns" here suggests that some ideal of language that is beyond individual expression. That language can be a formula or structure distinct from content, from materiality. This effectively turns expression into a *form*.

That there can be form without expression is a central fiction that drives the narrative of abundance. Text capture does not extract the "facts and rules of language" from language expression, rather, as I will demonstrate below, they sever the expression from its source. They take expression from the context of the one who has expressed it, for the sole purpose monetizing that severance. If ML-generated text is highly transformed, it is only because it is highly extracted, processed and therefore cleaned, like money laundered many times over.

1.4 web scraping

1.4.1 outline

- manipulating the HTTP protocol to mimic a browser request
 - HTTP protocol
 - from HTML -> python object
- parts of a URL
 - root, path, query
- bot blockers
 - requests are sometimes blocked
- bypassing blockers:
 - ghost browsers
 - scraping xhr
 - finding pages that can be scraped: congress.gov
- web crawling automates the data collection
 - crawling with scrapy: heritage.org/gender page

1.4.2 manipulating HTTP: Heritage Foundation

Websites are stored in web servers, accessed over a network. Web data travels over the network in the form of bytes, the smallest portable data unit. When the bytes reach the web browser, they are decoded into HTML language, which is assembled into a web page readable by human eyes.

Here is an example some the HTML on the "gender" topic page from *The Heritage Foundation*, on <https://Heritage.org/gender>. This section of the page shows the first three articles listed under the "gender" topic.

```
<a href="/gender/commentary/feminists-want-choice-only-when-it-the-right-one" class="result-card__link" title="article link"><span>
  Feminists Want Choice, but Only When It Is the Right One
</a>
```

```
<a href="/staff/jess-hamilton" class="result-card__link" title="article link"><span>
```

```

    Jess Hamilton
  </span>
</a><p class="result-card__date"><span>Aug 24, 2026</span>3 min
read</p>
<a href="/gender/commentary/the-quiet-way-gender-ideology-reaches-children"
class="result-card__title js-hover-target" title="article link">The
Quiet Way Gender Ideology Reaches Children</a>
<p class="result-card__date"><span>Jul 29, 2026</span> 3 min read</p>

<p class="result-card__eyebrow">First Principles</p>
<a href="/gender/report/title-ixs-failed-experiment-why-accommodating-sex-differences-l
title="heritage report" class="result-card__title
js-hover-target">Accommodating Sex Differences Beats Engineered
Parity</a>
    <a href="/staff/scott-yenor-phd" class="result-card__link">    <span>Scott
  </a>
  <p class="result-card__date"><span>Jul 8, 2026</span> 42 min read
</p>
<a href="/defense/commentary/military-effectiveness-should-be-determined-the-war-depart
class="result-card__title js-hover-target" title="article
link">Military Effectiveness Should Be Determined by The War
Department Not Judges</a>
<p class="result-card__date"><span>Jun 29, 2026</span> 4 min read </p>

```

Web scraping is the extraction of such HTML data from websites. This extraction is possible by manipulating HTTP, or HyperText Transfer Protocol, which is a wrapper language that encapsulates the bytes that travel over the network, like an envelope. This wrapper language stipulates strict instructions on the transit of bytes from the web server to the web browser. It not just the envelope, then, it is also detailed protocol for how the envelope may be opened.

A web scraper uses programming code to manipulate an HTTP request. Using HTTP protocol, it makes a request to a web server much like a user makes a request by typing in a URL into the browser. The data it gets back from the HTTP request is also the same as the data that comes to a browser after searching a URL—only the data is intercepted before it arrives at the browser, before it can be assembled into a web page.

Below is an example of the code used to scrape the home page of *The Heritage Foundation* website. The code uses the `bs4` and `requests` libraries

in python.²¹

```
from bs4 import BeautifulSoup
import requests
r = requests.get('https://www.heritage.org/gender')
source = r.content
soup = BeautifulSoup(source, 'html.parser')
```

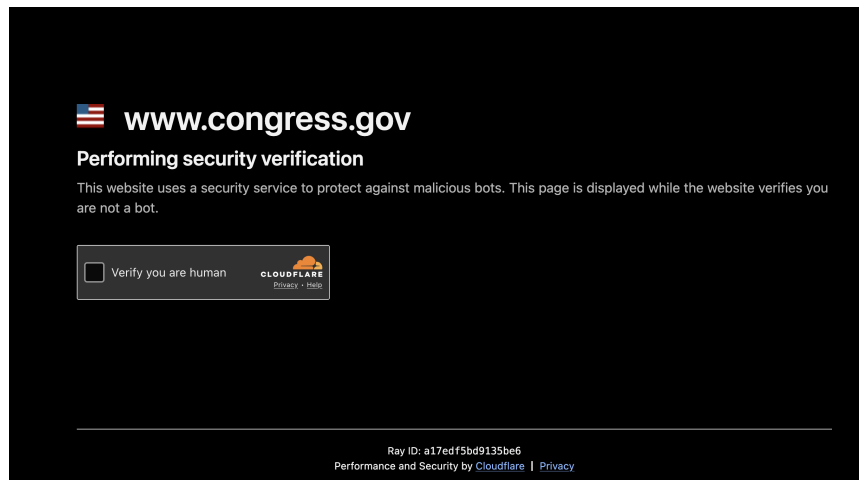
The `get()` function (on line 3) uses the HTTP method `GET` to request data from a server. The contents of request consist of a list of headers, identifying the requester, like "User-Agent," and certain properties about the request:

```
{
    'User-Agent': 'python-requests/2.32.3',
    'Accept-Encoding': 'gzip, deflate, br, zstd',
    'Accept': '*/*',
    'Date': 'Thu, 03 Sep 2026 12:21:09 GMT',
    'Content-Type': 'text/html; charset=UTF-8',
    'Transfer-Encoding': 'chunked',
    'Connection': 'keep-alive',
    'CF-Ray': 'a354b670ee7f41d9-EWR',
    'CF-Cache-Status': 'DYNAMIC',
    'Age': '5583',
    'Cache-Control':
    'max-age=10800, public',
    'Content-Language': 'en',
    'Expires': 'Sun, 19 Nov 1978 05:00:00 GMT',
    'Last-Modified': 'Thu, 03 Sep 2026 10:48:06 GMT',
    'Server': 'cloudflare',
    'Strict-Transport-Security': 'max-age=15552000; includeSubDomains; preload',
    'Vary': 'Accept-Encoding, Cookie, Cookie, Cookie',
    'Via': '1.1 varnish, 1.1 varnish, 1.1 varnish',
    'x-cache': 'HIT, HIT, MISS',
    'x-cache-hits': '12, 0, 0',
    'x-content-type-options': 'nosniff',
    'x-drupal-cache': 'MISS',
    'x-drupal-dynamic-cache': 'UNCACHEABLE (poor cacheability)',
```

²¹The `requests` library is one of the most popular libraries in the python programming language. Enabling programs to send and receive data through HTTP networks (i.e., the internet), it is the bedrock for virtually all web scraping and API tasks.

And as automated scrapers increase in frequency, so do the bot blockers.²² Over the past two years, Cloudflare, which runs on 20% of total internet pages and 50% of pages based in the US, has added increasing bot detection like "Challenges" and "Turnstiles" to its web pages.

Some of these operate under the surface and virtually invisible to the user, such as checking the HTTP request metadata like the "User-Agent." But beyond examining HTTP request data, bot blockers will use JavaScript code to check if the request is from a real web browser, rather than from an automated program. JavaScript, a programming language native to web browsers (discussed in more detail in Chapter 4: Text Display), can inject challenges for the browser to solve, challenges which may or may not involve the user. All of these challenges might require some processing from the browser, and increasingly, they may require the user's interaction to confirm that the request comes from a real browser, not a bot. One popular challenge is the "Security Verification" checkbox, which is increasingly visible across the internet (See Fig #).



1.4.5 bypassing blockers: ghost browsers

While these kinds of blockers are more advanced, there are ways of overcoming them. One method involves using a virtual or "ghost" browser through python libraries like `seleniumbase`.

²²Each year of the past few years, bot traffic has increased exponentially, coming mostly from major tech and AI companies, especially OpenAI, META, Google, and Anthropic. According to Cloudflare, 2025 saw increases of 19% in bot traffic (Cloudflare, "From Googlebot to GPTBot: who's crawling your site in 2025").

Instead of sending an HTTP request via a python script, `seleniumbase` routes the request through an actual web browser, one that is operated remotely by the python script. When the program is running, a browser window will open seemingly on its own. Depending on what content the user wants to scrape, it can then scroll, click on links, or fill in forms without the input of the user's mouse.

```
from seleniumbase import SB

with SB(uc=True, test=True, locale="en") as sb:
    url = "https://www.congress.gov/search?q=%7B%22congress%22%3A%5B%22119%22%5D%2C%22"
    sb.activate_cdp_mode(url)
    sb.sleep(2)
    sb.solve_captcha()
```

Because the program is using the actual browser window, it is able to trick the server into thinking it is a real, not automated, user. By waiting on the landing page for a few seconds, or even by clicking the checkbox, as indicated in the function `solve_captcha()` above, the JavaScript that triggers the "Security Verification" could thus be circumvented. With the checkpoint passed, the page loads normally. Meanwhile, the HTML data is sent to the python program running the request.

1.4.6 bypassing blockers: xhr

Besides HTML, there are other data streams, ones that load onto a web page after the HTML has loaded. Opposed to the *static* data of the HTML, which loads once and never changes, this kind of data is called *dynamic* data, and it loads continuously.

Dynamic data cannot be accessed by making a request to the page URL. Rather, the request must be made to another URL, which is encoded in the page scripts, which one can access in the "Network" tab in the browser developer tools.

The XHR, or XML HTTP Request can be used to gather search results, like the one pictured below. This request, which collects search results of the keyword "gender" from `foxnews.com`, gets the data from a server `moxie.foxnews.com`, where *Fox News* stores its database of article information.

```
headers = {
    'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:155.0) Gecko/20100
```

```

    'Accept': '*/*',
    'Accept-Language': 'en-US,en;q=0.9',
    'Referer': 'https://www.foxnews.com/',
    'Origin': 'https://www.foxnews.com',
    'Connection': 'keep-alive',
    'Sec-Fetch-Dest': 'empty',
    'Sec-Fetch-Mode': 'cors',
    'Sec-Fetch-Site': 'same-site',
  }

  params = {
    'q': 'gender',
    'fields': 'web',
  }

  response = requests.get('https://moxie.foxnews.com/search/aggregate/web', params=params)

```

If a ghost browser mimicks the user, then the XHR request thus mimicks a web page itself, feigning a request encoded in the page's JavaScript that would automatically load elements, like search results.

1.4.7 bypassing blockers: manipulating the URL

- hacking the URL

When dealing with a single website, there is always a work around.

Some pages on a website may block bots, while others do not. For example, the homepage and search results on **congress.gov** blocks all bots, unless the page is accessed through a ghost browser. But even without a ghost browser, some pages on the **congress.gov** server can be scraped, if one knows how to access them.

For example, the full text of congressional bills, which are not available through the official **congress.gov** API,²³ may be accessed by reverse engineering the bill URL. The bill URL for H.R.1927, the "End Taxpayer Funding of Gender Experimentation Act of 2021," proposed by the 117th Congress (2021-2022) can be accessed at the address below.

<https://www.congress.gov/bill/117th-congress/house-bill/1927>

²³Congress.gov API. <https://api.congress.gov/#/>

While the full text of this bill (which is under the "text" tab of the page) cannot be accessed without a ghost browser, it is also stored in another page on `congress.gov`.

`https://www.congress.gov/117/bills/hr1927/BILLS-117hr1927ih.htm`

The path of the URL is `/117/bills/hr1927/BILLS-117hr1927ih.html`, in contrast to the one above, which is `/bill/117th-congress/house-bill/1927`. This `/117` path, unlike the `/bill` one, can be scraped using traditional means.

```
r = requests.get("https://www.congress.gov/117/bills/hr1927/BILLS-117hr1927ih.htm")
```

The contents of the request contain the following text, showing the header of the bill.

```
<html><body><pre>
[Congressional Bills 117th Congress]
[From the U.S. Government Publishing Office]
[H.R. 1927 Introduced in House (IH)]

<DOC>
```

```
117th CONGRESS
  1st Session
```

```
H. R. 1927
```

```
To prohibit taxpayer-funded gender reassignment medical interventions,
and for other purposes.
```

This URL has valuable information that would enable one to scrape data from more bills. With access to another URL for comparison, the a formula for crafting more URLs could be reverse engineered.

For example, a URL for a different bill, H.R. 8731, "Protect Children's Innocence Act," also from the 117th congress (2021-2022) has a similar path to the former bill. Looking at both URLs side by side displays a pattern in the path's logic.

```
https://www.congress.gov/117/bills/hr8731/BILLS-117hr8731ih.htm
https://www.congress.gov/117/bills/hr1927/BILLS-117hr1927ih.htm
```

The differences between the two URLs reveals that the bill number, 8731 and 1927 is the only element that changes. With that knowledge, and with a list of relevant bill numbers (either downloaded from `congress.gov` search results or via the API), one can reverse engineer many—hundreds, in this case—of URLs for scraping.

```
https://www.congress.gov/117/bills/hr1927/BILLS-117hr1927ih.htm
https://www.congress.gov/117/bills/hr8731/BILLS-117hr8731ih.htm
https://www.congress.gov/117/bills/hr5/BILLS-117hr5ih.htm
https://www.congress.gov/117/bills/hr7684/BILLS-117hr7684ih.htm
https://www.congress.gov/117/bills/hr1475/BILLS-117hr1475ih.htm
https://www.congress.gov/117/bills/hr8170/BILLS-117hr8170ih.htm
https://www.congress.gov/117/bills/hr2695/BILLS-117hr2695ih.htm
https://www.congress.gov/117/bills/hr9497/BILLS-117hr9497ih.htm
https://www.congress.gov/117/bills/hr4176/BILLS-117hr4176ih.htm
https://www.congress.gov/117/bills/hr5744/BILLS-117hr5744ih.htm
```

Below is a function that passes a list of bill numbers into an algorithm that plugs the number into the URL and scrapes it. The resulting dataset contains the text for 688 individual bills.

```
def scrape_bill_text(bill_numbers):
    bills_text = []
    for number in bill_numbers:
        url = (f'https://www.congress.gov/117/bills/hr{number}/BILLS-117hr{number}ih.htm')
        response = requests.get(url)
        content = response.content
        bills_text.append(content)
    return bills_text
```

1.4.8 web crawling: heritage.org

There is another thing: Web crawling.

The automation of scraping multiple pages is called "web crawling." Traditionally, web crawling is an accretative procedure that involves scraping a page, gathering the links for that page, and scraping those pages.

For example, the following web crawler scrapes the search results from the `Heritage.org/gender` topic page. The crawler has two functions: `parse()`,

which scrapes search results, and `parse_article()`, which scrapes article content linked from the search results.

```
def parse(self, response):
    article_page_links = response.css("div.result-card__info-wrapper a::attr('href')")
    yield from response.follow_all(article_page_links, self.parse_article)

    # Go to the next page of results
    next_page = response.css('li.pager__item a::attr("href")').get()
    if next_page is not None:
        yield response.follow(next_page, self.parse)
```

In this function, two important actions are indicated. First, the `yield` statement contains another function, `follow_all()`, which first grabs all of the links embedded on that page. Then, `follow_all()` triggers another function, `parse_article()`, which grabs the article text from the linked page. The second important action is what happens next, which is pagination. The `parse()` function will then grab the link for the next page of results, grab all the result information from that page, and trigger the `parse_article()` function again.

The process then repeats itself, over and over again, until all pages of results have been scraped. Crawling is thus a recursive process.

1.5 TO_DRAFT close reading: heritage foundation

1.5.1 scaling small

It seems that there is a corruption that is inherent to scale, a problem first pointed out in Bender et al's titular question, "Can Language Models be too Big?". As Samuel A. Moore writes, "Scalability banishes meaningful diversity, that is, diversity that might change things" (Samuel A. Moore, *Publishing Beyond the Market* 147).

Like with the Zyte report, there is a cycle here in which increasing data and AI technology feed each other. This cycle manifests in real circular relationships between companies, where AI companies who have the technology must work with cloud companies who have the necessary servers and GPUs to handle higher compute. The vast majority of the capital raised by these AI companies are paid to the cloud providers, who in turn are responsible for majority of the investment into the AI companies. As the authors point out, "These circular investment deals can blur the boundary between 'investment' and 'acquisition'". (Varoquaux et al 8).

1.6 TO_DRAFT conclusion: shame, sedgwick, contagion

1.6.1 shame as contagion

Shame thus moves outward, creating a nexus for interpersonal relation. Sedgwick links shame to contagion:

“Shame—living, as it does, on and in the muscles and capillaries of the face—seems to be uniquely contagious from one person to another. And the contagiousness of shame is only facilitated by its anamorphic, protean susceptibility to new expressive grammars” (*Touching Feeling* 63).

“not a discrete intrapsychic structure, but a kind of free radical that (in different people and different cultures) attaches to and permanently intensifies or alters the meaning of—of almost anything: a zone of the body, a sensory system, a prohibited or indeed a permitted behavior, another affect such as anger or arousal, a named identity, a script for interpreting other people’s behavior toward oneself” (*Touching Feeling* 62)

In the orbit of queerness, where negative experiences are identity forming, Sedgwick looks at shame as a positive force.

Shame interests me politically, then, because it generates and legitimates the place of identity—the question of identity... without giving that identity space the standing of an essence. It constitutes the as-to-be-constituted, which is also to say, as already there for the (necessary, productive) misconstrual and misrecognition. (*Touching Feeling* 63)

1.6.2 Sedgwick’s critique of Butler

In the influential final pages of *Gender Trouble*, for example, Butler offers a programmatic argument in favor of demystification as “the normative focus for gay and lesbian practice,” with such claims as that “drag implicitly reveals the imitative structure of gender itself” (137); “we see sex and gender denaturalized by means of a performance” (138); “gender parody reveals that the original identity... is an imitation” (138); “gender performance will enact and reveal the performativity of gender itself” (139); “parodic repetition... exposes the phantasmatic effect of abiding identity” (141); “the parodic repetition of gender exposes... the illusion of gender identity” (146); and

"hyperbolic exhibitions of 'the natural' . . . reveal its fundamentally phantasmatic status" (147) as well as "exposing its fundamental unnaturalness" (149; all emphases added). (*Novel Gazing* 17).